RoboCup MSL
Scientific challenge 2022



T. Kergourlay - Q. Dominiak - A. H. Elsayed - A. Bassiouny - Z. El Bannoudi - M. Dhainaut - A. A. Khan - A. Meaudre - F. Albrand - X. Chéreau -T. Wemaere - M. Jamme - A. Kaibaldiyev - C. Bohler - Y. Masinski - V. Barchasz - T. Soriano - J. Quilles (robot-maker.com)

Presenter : V. Gies

UNIVERSITÉ DE TOULON

RoboCup

Robust absolute positioning using SLAM and Kalman Filtering

- Absolute positioning without a soccer field : why and how ?
- Lidar SLAM (Simultaneous Location and Mapping) : a robust absolute positioning method for MSL
  - SLAM Algorithm
  - Results
- Coupling SLAM and Kalman filtering : a robust and accurate positioning method for MSL
  - Implementation
  - Results



RoboCup MSL - Scientific challenge 2022

Robust absolute positioning using SLAM and Kalman Filtering

- Absolute positioning without lines : why and how ?
  - In a real soccer field, lines are hard to see.
    - Goal posts can be used
    - Field boundary can be used
  - Want to be able to play anywhere !
    - Simplifying tests
    - Simplifying demo (without any equipment : no lines / no goal)



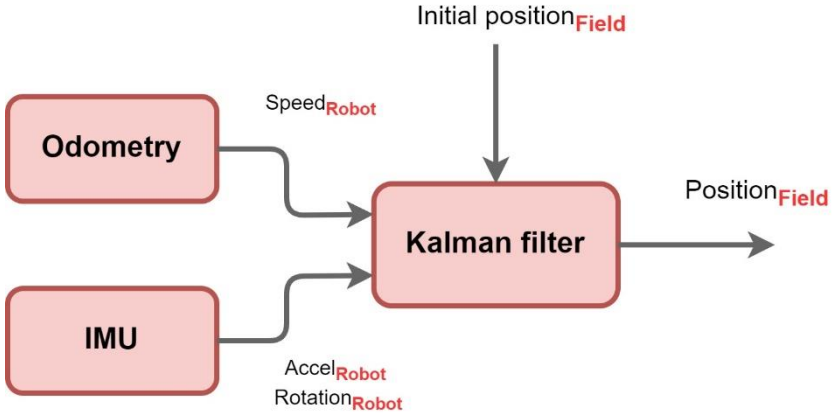
- Last year : proposed to use camera for detecting posts
  - Some issues
    - Using deep learning : latency
    - Distance is not accurate : +/- 2m at 8m
    - Finding small elements is sometimes difficult

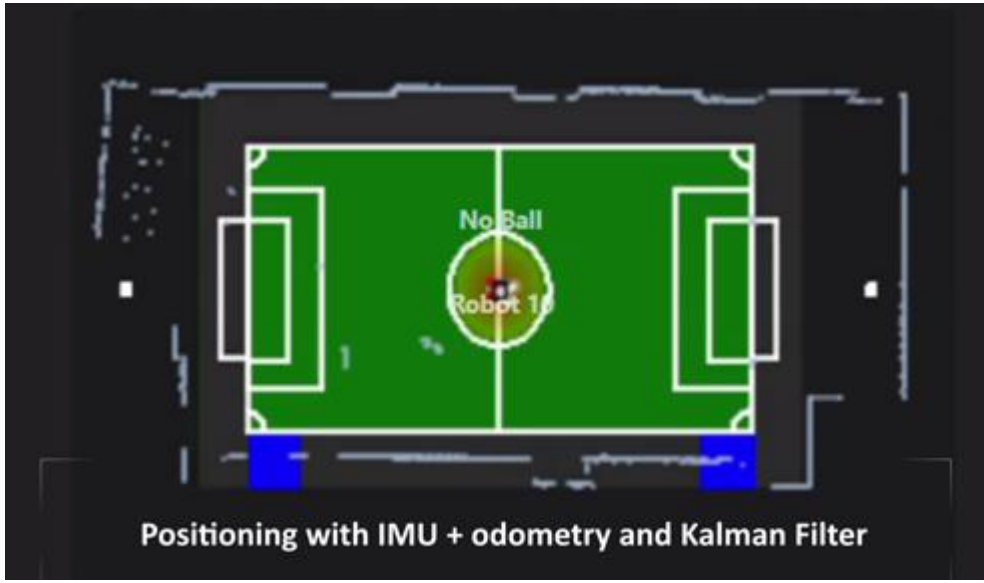


Robust absolute positioning using SLAM and Kalman Filtering

UNIVERSITÉ DE TOULON

- What is possible without absolute positioning ?
  - Like running in an infinite white room with nothing inside
  - Basic version : IMU + odometry





  - Issue : integrating offsets
    - Sensor offsets
    - Mechanical slacks
    - Sliding
  - Absolute positioning is really necessary

Robust absolute positioning using SLAM and Kalman Filtering

- Absolute positioning without a soccer field : why and how ?
- Lidar SLAM (Simultaneous Location and Mapping) : a robust absolute positioning method for MSL
  - SLAM Algorithm
  - Results
- Coupling SLAM and Kalman filtering : a robust and accurate positioning method for MSL
  - Implementation
  - Results



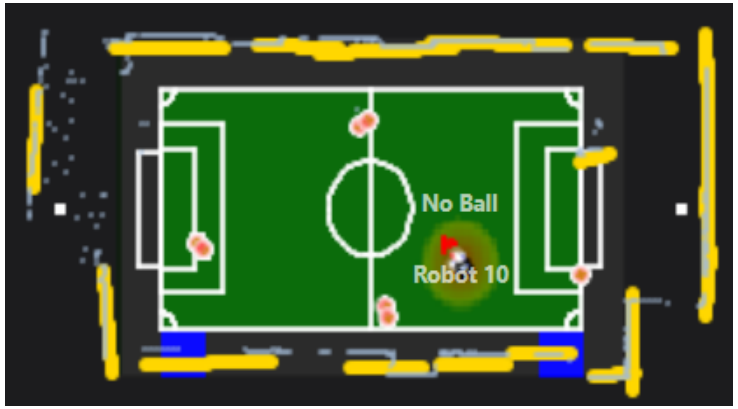
RoboCup MSL - Scientific challenge 2022

Robust absolute positioning using SLAM and Kalman Filtering

- Lidar SLAM (Simultaneous Location and Mapping) : a robust positioning method for MSL
  - Using lidar
  - Improved performance :
    - Accuracy : positioning at $\sigma_X = \sigma_Y = 15$cm in any room
    - Robustness : always positioning (no occlusions)
    - Low latency : less than 2ms
      - Compared with camera deep learning
    - Reduced computational cost
      - Compared with camera deep learning



- A step toward 2050 objectives
  - Playing on any field material
  - Improved safety
    - Positioning reliably robots / humans



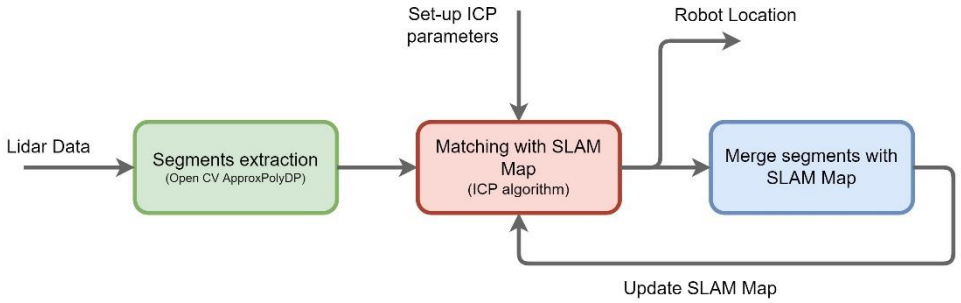
Robust absolute positioning using SLAM and Kalman Filtering

UNIVERSITÉ DE TOULON

SLAM Algorithm

Iteration :

- Segment List : extraction of segment list from lidar points
  - Using OpenCV
- Matching : Lidar Segment List vs SLAM Map
  - Using Iterative Closest Points (ICP) algorithm on segments
  - Finds the new location
- Updating Map
  - Merging segment list in SLAM map
  - Forgetting rarely seen segments in SLAM map



SLAM Algorithm

ICP Problem formulation
Given 2 sets of segments

$$\{p_i\}, \{q_i\}, i = 1..n \text{ in } \mathbb{R}^3$$

Find the rigid transform R *and* q_0 *minimizing :*

$$\sum_{i=1}^{n} \|R(q_i) + q_0 - p_i\|^2$$



Robust absolute positioning using SLAM and Kalman Filtering

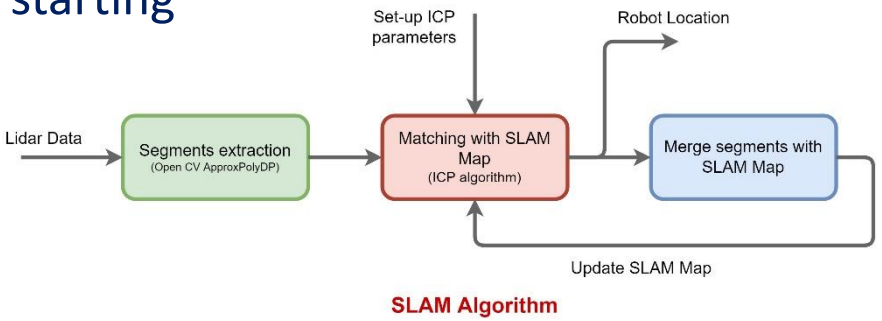
SLAM Algorithm : Results

- SLAM Precision : $\sigma_X = \sigma_Y = 15cm$
- Real time :
  - Computed at 50 Hz (every 20ms)
  - Less 1% of CPU (i7 9700)
    - ΔX / ΔY searching range : 0.2m
      - Robot max speed : 10 m/s
    - Δθ searching range : 15°
  - Map size :
    - 20-30 segments
- Robustness
  - Robust to occlusions due to robots, humans, and small objects

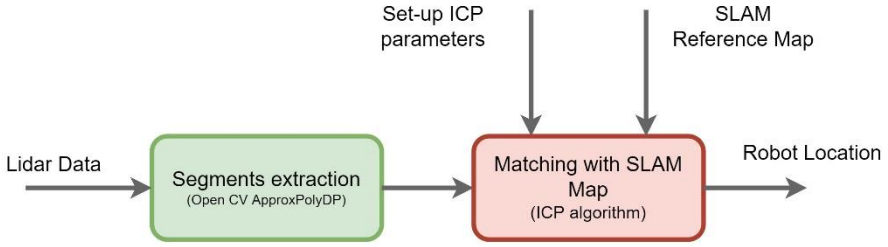


Robust absolute positioning using SLAM and Kalman Filtering

- Using Lidar SLAM in operational conditions
  - Set-up : generate SLAM reference map starting from position (X=0, Y=0, θ=0)
    - Only one time for all robots
  - During game : positioning using SLAM reference map
    - Initial positioning
      - Full Search ICP
        - Takes more time (0.1 seconds on a i7 – 9700 CPU)
      - Or push « reset position » in a defined location
    - Computation of position :
      - Limited search ICP
        - Very light : less 1% of CPU (i7 9700)
        - Low latency : 2ms



SLAM Algorithm



Positioning Algorithm using SLAM Reference Map

Robust absolute positioning using SLAM and Kalman Filtering

- Absolute positioning without a soccer field : why and how ?
- Lidar SLAM (Simultaneous Location and Mapping) : a robust absolute positioning method for MSL
  - SLAM Algorithm
  - Results
- Coupling SLAM and Kalman filtering : a robust and accurate positioning method for MSL
  - Implementation
  - Results

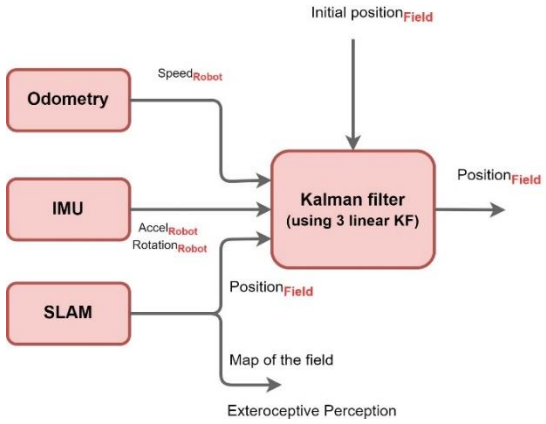


RoboCup MSL - Scientific challenge 2022

Robust absolute positioning using SLAM and Kalman Filtering

UNIVERSITÉ DE TOULON

- Coupling SLAM and Kalman filtering - a robust and accurate positioning method for MSL
  - Coupling 3 linear Kalman filters :
    - Kalman for estimating θ :
      - Inputs : IMU : Gyro V_θ - Odometry : V_θ - SLAM positioning : θ
    - Kalman for estimating V_x / V_y :
      - Inputs : IMU : Accel X/Y - Odometry : V_x / V_y
    - Kalman for estimating X/Y :
      - Inputs : Kalman V_x / V_y - SLAM positioning : X / Y
    - Better than using a single extended Kalman filter



Predict

Predicted state estimate $\hat{\mathbf{x}}_{k|k-1} = \mathbf{F}_k\hat{\mathbf{x}}_{k-1|k-1} + \mathbf{B}_k\mathbf{u}_k$

Predicted error covariance $\mathbf{P}_{k|k-1} = \mathbf{F}_k\mathbf{P}_{k-1|k-1}\mathbf{F}_k^T + \mathbf{Q}_k$

Correct

Innovation residual $\tilde{\mathbf{y}}_k = \mathbf{z}_k - \mathbf{H}_k\hat{\mathbf{x}}_{k|k-1}$

Innovation covariance $\mathbf{S}_k = \mathbf{H}_k\mathbf{P}_{k|k-1}\mathbf{H}_k^T + \mathbf{R}_k$

Optimal Kalman gain $\mathbf{K}_k = \mathbf{P}_{k|k-1}\mathbf{H}_k^T\mathbf{S}_k^{-1}$

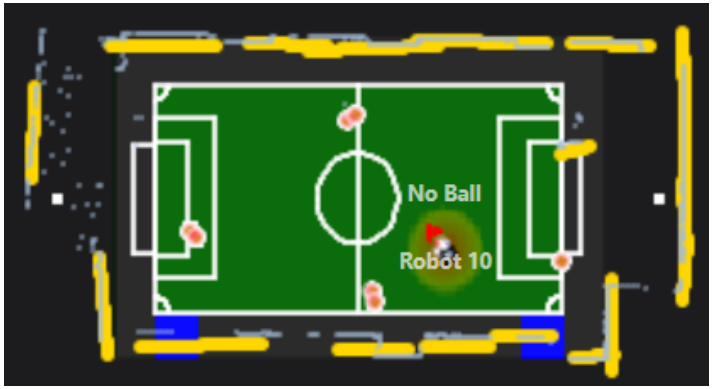
Corrected state estimate $\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k\tilde{\mathbf{y}}_k$

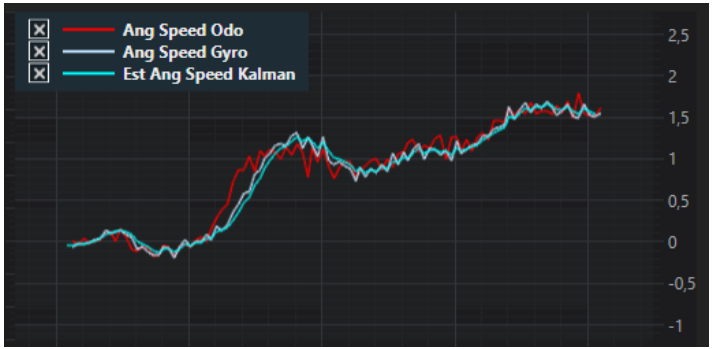
Corrected estimate covariance $\mathbf{P}_{k|k} = (\mathbf{I} - \mathbf{K}_k\mathbf{H}_k)\mathbf{P}_{k|k-1}$



Robust absolute positioning using SLAM and Kalman Filtering

- Coupling SLAM and Kalman filtering : results
  - Improved performance :
    - Accuracy : positioning at $\sigma_X = \sigma_Y = 4$cm in any room without field elements
    - Robustness : always positioning
    - Real time :
      - Computed at 50 Hz (every 20ms)
      - Low latency : less than 2ms
    - Low computational cost :
      - Less 2% of CPU (i7 9700)
  - Fully operational in RCT robots
    - Theoretically complex : more than 1 year of work
    - Validated with 3 robots and 5 humans
    - Publication in progress





- Conclusion :
  - A contribution to the MSL community
    - A scientific paper will describe in details concepts and implementations
    - Teams are welcome to ask for a support for implementing SLAM and Kalman filters.
  - A major step for our team
    - Playing without absolute positioning was not reliable
    - With Lidar SLAM, Kalman filtering, robots behaviour is stable and is exactly the same as in our simulator
      - Open interesting perspectives for the next years
  - Perspectives
    - Coupling to Visual SLAM using cameras

References :

- Debeunne, César & Vivet, D.. (2020). A Review of Visual-LiDAR Fusion based Simultaneous Localization and Mapping. Sensors. 20. 2068. 10.3390/s20072068.
- Khan, Misha Urooj & Zaidi, Syed Azhar Ali & Ishtiaq, Arslan & Bukhari, Syeda & Samer, Sana & Farman, Ayesha. (2021). A Comparative Survey of LiDAR-SLAM and LiDAR based Sensor Technologies. 10.1109/MAJICC53071.2021.9526266.
- Elias Maia Azevedo & Dias Ferreira (2021) Improving Lidar Odometry and Mapping in Real-time using Inertial Measurements
- Kuramachi, Ryo & Ohsato, Akihito & Sasaki, Yoko & Mizoguchi, Hiroshi. (2015). G-ICP SLAM: An odometry-free 3D mapping system with robust 6DoF pose estimation. 176-181. 10.1109/ROBIO.2015.7418763.

Thanks for your attention
Questions ?



Welcome to France in 2023 !